VFX 705

PROJECT 02

ANIMATED PIXEL ART QUILT

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INTRODUCTION

Each motif consists of six animated frames. Compared to the WIP version, this final iteration refines visual details, improves color consistency, and enhances overall clarity.

FRAME_01



FRAME_02



FRAME_03



FRAME_04



FRAME_05



FRAME_06



GET DATA FROM FILES

01 READ ALL THE FILE

Read each .c file and extract the frame settings (frame count, width, and height) together with all hexadecimal color values.

02 SAVE ALL THE HEX IN ONE LIST

Store every hex color from all files into a single list to prepare for structured frame reconstruction and later color processing.

03 TAKE VALUES FROM THE LIST AND ASSIGN TO KEYS IN A DICTIONARY

Group the collected hex values by tile name and organize them into frame-based pixel data using a dictionary structure.

04 CHANGE HEX TO RGB AND GET COLOR PALETTE

Convert hex colors to normalized RGB values and generate a unique color palette for each animated tile.

USER GUIDE

1



The quilt layout can be adjusted by modifying the cols and rows values, which control the number of tiles arranged horizontally and vertically.

2

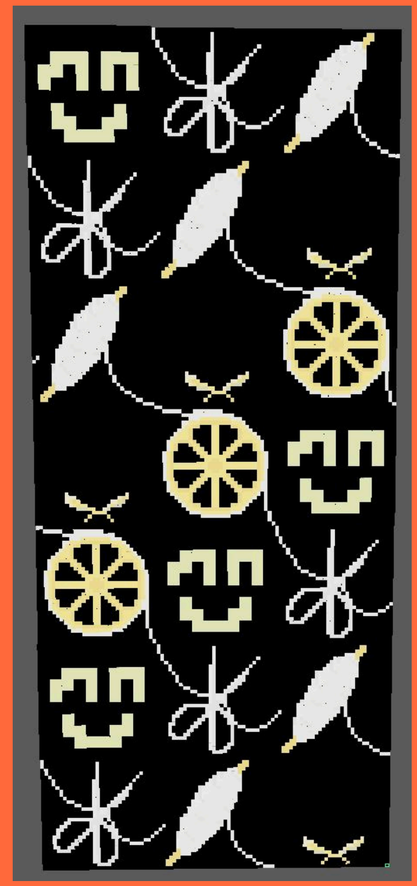
```
tile_palette_dict[tile] = palette
```

```
dict_keys(['olive', 'clew', 'sissor', 'laurel', 'wheel'])
```

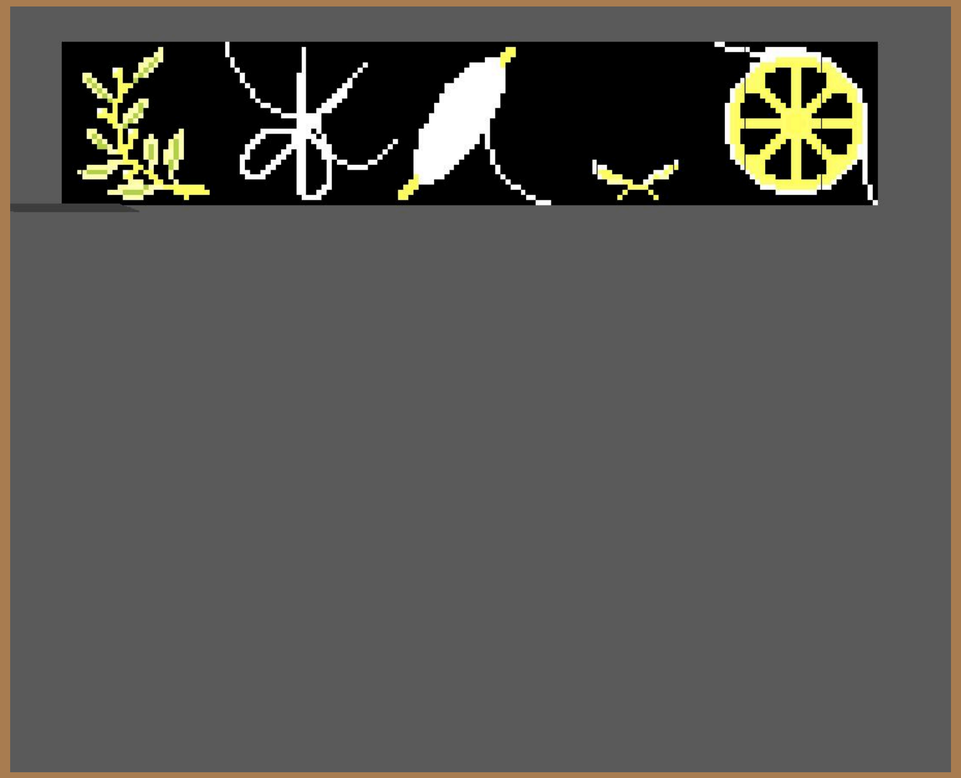
Each tile uses a single shared color palette that is generated from the union of colors across all its animation frames.

3

The read_file() function parses each .c file independently, extracting frame count, width, height, and hex pixel data. It then splits the pixels into correctly sized frames, allowing the script to support .c files with different resolutions without breaking layout or indexing.



PROBLEM ENCOUNTERED DURING ANIMATION BUILD



During the execution of the visibility-animation function, the generated pattern consistently appeared as only a single row instead of a full animated tile.
After debugging the loop structure, I discovered that the issue was caused by the return statement being placed inside the loop.
Because return immediately exits the function, the function stopped after completing only the first iteration, preventing the remaining frames and pixels from being created.

FUTURE IMPROVEMENTS

- Could store and use the original frame width and height from each .c file instead of inferring square dimensions from pixel counts. This would allow the quilt system to fully support non-square resolutions and improve spatial accuracy.
- Could support tiles with different frame values by computing the lowest common multiple of all frame counts to create a unified loop length. Each tile would repeat using modulo indexing, allowing the entire quilt animation to loop seamlessly.

UPDATED VERSION

- Moved all constants (PFX, tileSize, cols, rows, file paths) to the top of the script for better organization.
- Added a single top-level group (LZ_quilt_grp) that parents all tile groups to ensure a clean hierarchy structure.
- Optimized file parsing by unpacking read_file() results in a single call instead of multiple calls.

```
for curr_file in file_names:
    full_path = os.path.join(file_path, curr_file)
    #delete the ".c"
    name = curr_file.replace(".c", "")
    #how many pixels in one file
    fcount = read_file(full_path)[0]
    w = read_file(full_path)[1]
    h = read_file(full_path)[2]
    frames = read_file(full_path)[3]

    whole_dict[name] = frames
```

